

key abstractions, the most important of which are processes and threads, address spaces, and files. Accordingly the next three chapters are devoted to these topics.

Chapter 2 is about processes and threads. It discusses their properties and how they communicate with one another. It also gives a number of detailed examples of how interprocess communication works and how to avoid some of the pitfalls.

In Chap. 3 we study address spaces and their adjunct, memory management. The important topic of virtual memory will be examined, together with paging.

Then, in Chap. 4, we come to the topic of file systems. To a considerable extent, what the user sees is the file system. We will look at both the file-system interface and the file-system implementation.

Input/Output is covered in Chap. 5. We will cover the concept of device (in)dependence using examples such as storage devices, keyboards, and displays.

Chapter 6 is about deadlocks, including ways to prevent or avoid them.

At this point, we will have completed our study of the basic principles of single-CPU operating systems. However, there is more to say, especially about advanced topics. In Chap. 7, we examine virtualization. We discuss both the principles, and some of the existing virtualization solutions in detail. Another advanced topic is multiprocessor systems, including multicores, parallel computers, and distributed systems. These subjects are covered in Chap. 8. Another important subject is operating system security, which we cover in Chap 9.

Next we have some case studies of real operating systems. These are UNIX, Linux, and Android (Chap. 10), and Windows 11 (Chap. 11). The text concludes with some wisdom and thoughts about operating system design in Chap. 12.

1.11 METRIC UNITS

To avoid any confusion, it is worth stating explicitly that in this book, as in computer science in general, metric units are used instead of traditional English units (the furlong-stone-fortnight system). The principal metric prefixes are listed in Fig. 1-31. The prefixes are typically abbreviated by their first letters, with the units greater than 1 capitalized. Thus a 1-TB database occupies 10^{12} bytes of storage and a 100-psec (or 100-ps) clock ticks every 10^{-10} seconds. Since milli and micro both begin with the letter “m,” a choice had to be made. Normally, “m” is for milli and “ μ ” (the Greek letter mu) is for micro.

It is also worth pointing out that, in common industry practice, the units for measuring memory sizes have slightly different meanings. There kilo means 2^{10} (1024) rather than 10^3 (1000) because memories are always a power of two. Thus a 1-KB memory contains 1024 bytes, not 1000 bytes. Similarly, a 1-MB memory contains 2^{20} (1,048,576) bytes and a 1-GB memory contains 2^{30} (1,073,741,824) bytes. However, a 1-Kbps communication line transmits 1000 bits per second and a 1-Gbps LAN runs at 1,000,000,000 bits/sec because these speeds are not powers of two. Unfortunately, many people tend to mix up these two systems, especially for

Exp.	Explicit	Prefix	Exp.	Explicit	Prefix
10^{-3}	0.001	milli	10^3	1,000	Kilo
10^{-6}	0.000001	micro	10^6	1,000,000	Mega
10^{-9}	0.000000001	nano	10^9	1,000,000,000	Giga
10^{-12}	0.0000000000001	pico	10^{12}	1,000,000,000,000	Tera
10^{-15}	0.0000000000000001	femto	10^{15}	1,000,000,000,000,000	Peta
10^{-18}	0.0000000000000000001	atto	10^{18}	1,000,000,000,000,000,000	Exa
10^{-21}	0.0000000000000000000001	zepto	10^{21}	1,000,000,000,000,000,000,000	Zetta
10^{-24}	0.000000000000000000000001	yocto	10^{24}	1,000,000,000,000,000,000,000,000	Yotta

Figure 1-31. The principal metric prefixes.

SSD or disk sizes. To avoid ambiguity, in this book, we will use the symbols KB, MB, and GB for 2^{10} , 2^{20} , and 2^{30} bytes, respectively, and the symbols Kbps, Mbps, and Gbps for 10^3 , 10^6 , and 10^9 bits/sec, respectively.

1.12 SUMMARY

Operating systems can be viewed from two viewpoints: resource managers and extended machines. In the resource-manager view, the operating system's job is to manage the different parts of the system efficiently. In the extended-machine view, the system provides the users with abstractions that are more convenient to use than the actual machine. These include processes, address spaces, and files.

Operating systems have a long history, starting from the days when they replaced the operator, to modern multiprogramming systems. Highlights include early batch systems, multiprogramming systems, and personal computer systems.

Since operating systems interact closely with the hardware, some knowledge of computer hardware is useful to understanding them. Computers are built up of processors, memories, and I/O devices. These parts are connected by buses.

Operating systems can be structured as monolithic, layered, microkernel/client-server, virtual machine, or exokernel/unikernel systems. Regardless, the basic concepts on which they are built are processes, memory management, I/O management, the file system, and security. The main interface of an operating system is the set of system calls that it can handle. These tell us what it really does.

PROBLEMS

1. What are the two main functions of an operating system?
2. What is multiprogramming?